Lab 7: Power Analysis: Part 1

ECE2200L Section 04, Fall 2026

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No 1.

Step 1. Determine the expression for the instantaneous power given that both voltage and current are sinusoidal functions given below

$$v(t) = V_m \cos(\omega t + \theta_v), \quad i(t) = I_m \cos(\omega t + \theta_i)$$

$$p(t) = v(t)i(t)$$

$$p(t) = V_m I_m \cos(\omega t + \theta_v)\cos(\omega t + \theta_i)$$

recall that

$$\cos(A)\cos(B) = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

thus

$$\boxed{p(t) = \frac{V_m I_m}{2}[\cos(\theta_v - \theta_i) + \cos(2\omega t + \theta_v + \theta_i)]}$$

Step 2. Sketch: $v(t)$, $i(t)$, and power $p(t)$ all on the same graph

View simulation code

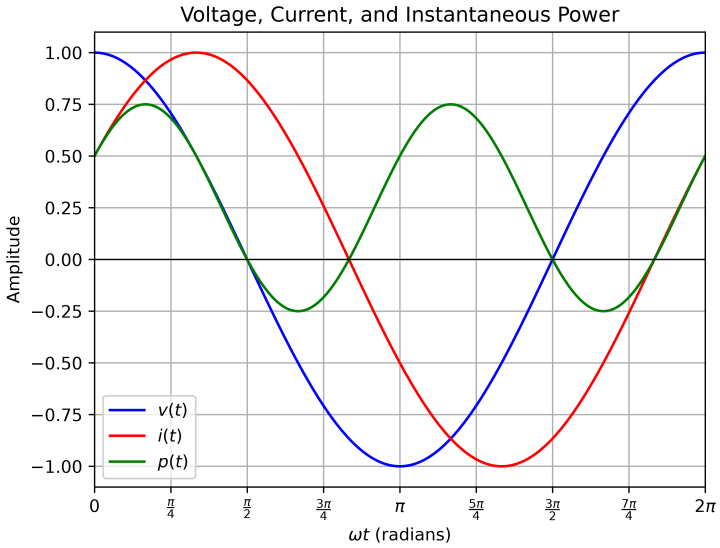


Figure 1: Sinusoidal voltage $v(t)$, current $i(t)$, and instantaneous power $p(t)$ plotted over one complete cycle, using assumed values $V_m = 1\,\text{V}$, $I_m = 1\,\text{A}$, $\omega = 1\ \text{rad/s}$, $\theta_v = 0°$, and $\theta_i = -60°$. The current lags the voltage by $60°$, while the instantaneous power oscillates at twice the frequency of the voltage and current.

No 2.

Step 1. Suppose the power factor of an impedance is 0.7 leading. What is the simplest model of this impedance?

A leading power factor means current leads voltage, so the impedance angle is negative:

$$\varphi = -\cos^{-1}(0.7) = -45.57^\circ$$

The impedance consists of some (positive) resistance and some (negative) reactance, so a resistor and capacitor in series can represent it.

Step 2. Simulate the circuit you designed in Step 1 to confirm your design.

assuming $R = 100\,\Omega$:

$$Z = R + \frac{1}{j\omega C} = R + jX_C$$

$$\tan(\varphi) = \frac{\Im(Z)}{\Re(Z)} = \frac{X_C}{R}$$

$$\Rightarrow |X_C| = R\tan|\varphi| \approx 102.02\,\Omega$$

assuming a frequency of 60 Hz,

$$|X_C| = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$$C \approx 26\,\mu\text{F}$$

View simulation code

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C = 26.001 uF
Z = 100.000 - j102.020 ohm
Power factor = 0.7000
Current leads voltage by 45.57 degrees
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No 3.

Step 1. Find the average power, the reactive power, and the power factor for the circuit shown.

$$i(t) = 20\cos(\omega t + 15^\circ)\ \text{A}, \quad v(t) = 100\cos(\omega t - 45^\circ)\ \text{V}$$

$$\mathbf{I} = 20\angle 15^\circ, \quad \mathbf{V} = 100\angle -45^\circ$$

$$\mathbf{S} = \frac{1}{2}\mathbf{V}\mathbf{I}^* = \frac{1}{2}(100\angle -45^\circ)(20\angle -15^\circ) = 1000\angle -60^\circ$$

$$\Rightarrow \mathbf{S} = 500 - j866.025\ \text{VA}$$

$$\Rightarrow \boxed{P = 500\,\text{W}, \quad Q = -866.025\ \text{VAR}}$$

$$\varphi = \arg(\mathbf{I}) - \arg(\mathbf{V}) = 60^\circ \text{ (leading)}$$
$$|\cos(\varphi)| = 0.5$$
$$\Rightarrow \boxed{\text{pf} = 0.5 \text{ leading}}$$

Step 2. Is the power factor leading or lagging?

$$\arg(\mathbf{I}) - \arg(\mathbf{V}) > 0 \Rightarrow \text{leading}$$

Step 3. Estimate the real and imaginary parts of the power

$$\Re(\mathbf{S}) = P = 500\,\text{W}$$
$$\Im(\mathbf{S}) = Q = -866.025\ \text{VAR}$$

No 4.

Step 1. Find the average power, the reactive power, and the power factor for the circuit shown.

$$i(t) = 20\cos(\omega t + 165^\circ)\ \text{A}, \quad v(t) = 100\cos(\omega t - 45^\circ)\ \text{V}$$

$$\mathbf{I} = 20\angle 165^\circ, \quad \mathbf{V} = 100\angle -45^\circ$$

$$\mathbf{S} = \frac{1}{2}\mathbf{V}\mathbf{I}^* = 1000\angle -210^\circ \equiv 1000\angle 150^\circ$$

$$\Rightarrow \mathbf{S} = -866.025 + j500\ \text{VA}$$

$$\Rightarrow \boxed{P = -866.025\,\text{W}, \quad Q = 500\ \text{VAR}}$$

$$\varphi = \arg(\mathbf{I}) - \arg(\mathbf{V}) = 210^\circ \equiv 150^\circ \text{ (lagging)}$$
$$|\cos(\varphi)| = 0.866$$
$$\Rightarrow \boxed{\text{pf} = 0.866 \text{ lagging}}$$

Step 2. Estimate what the simplest model of the circuit inside the box is, and why?

$$P < 0 \Rightarrow \text{delivering real power}$$
$$Q > 0 \Rightarrow \text{absorbing reactive power}$$

$$Z = \frac{\mathbf{V}}{\mathbf{I}} = 5\angle -210^\circ = 5\angle 150^\circ$$
$$Z = -4.33 + j2.5\Omega$$

Thus, the simplest model that can represent the circuit is an inductor and an active element/source in series. It should have an equivalent resistance of $-4.33\,\Omega$ and an equivalent reactance of $2.5\,\Omega$.